



LIYA NIU

Phone: +86 135 8152 1040
Email: lyniu25@stu.pku.edu.cn
Address: Physics Building, Peking
University, 209 Chengfu Road, Haidian
District, Beijing, China
English proficiency: CET-4 563; CET-6 576



Research Interests: Climate engineering; Solar radiation modification; Marine Cloud Brightening (MCB); aerosol-cloud interactions; low clouds; satellite remote sensing; cloud radiative effects

EDUCATION

Sep 2025 - Present	Peking University	Atmospheric Physics and Atmospheric Environment	Doctor of Philosophy
Sep 2021 - Jun 2025	China Agricultural University	Applied Meteorology	Bachelor of Science

SELECTED RESEARCH PROJECTS

- 1. Occurrence Characteristics and Future Projections of Compound Extreme Hot-Dry Wind Events in the North China Winter Wheat Region**
Sep 2022 - Jun 2025 | Principal Investigator
Beijing Municipal Undergraduate Research Project.
- 2. Regional Differences, Causes, and Countermeasures of Climate Warming in China**
Sep 2023 - Jan 2024 | Project Lead
Youth Science and Technology Green Innovation Competition Project.
- 3. Theory and Technology of Low-Cloud Radiative Intervention Geoengineering and Its Climate Effects**
Dec 2024 - Nov 2029 | Major Participant
National Key Research and Development Program of China.

HONORS AND AWARDS

- Outstanding Communist Party Member, School of Physics, Peking University (Mar 2026)
- Beijing Outstanding Student and Beijing Outstanding Graduate (Jun 2025)
- Hello Campus Dream-Building Scholarship (Jun 2024)
- Embry Form Scholarship (Jun 2022)
- Outstanding Student and First-Class Academic Scholarship, China Agricultural University (awarded for three consecutive years)

PUBLICATIONS AND MANUSCRIPTS

1. Niu Liya, Zhao Chuanfeng*, Zhang Haotian, et al. 2026. Spatiotemporal Characteristics of Aerosols and Low Clouds from MODIS and Their Implications for Marine Cloud Brightening. Science China Earth Sciences. [SCI Q1 Top]
2. Zhang Haotian, Zhao Chuanfeng*, Chen Annan, Niu Liya, et al. 2026. Impacts of Stratospheric Aerosol Injection Geoengineering Deployment and Termination on Extreme Climate Events in China. Science China Earth Sciences. [SCI Q1 Top]
3. Zhao Chuanfeng*, Niu Liya, Li Jiefeng, Zhang Haotian. 2026. Marine cloud brightening: state-dependent efficacy, deployment design, and systemic risks. Earth-Science Reviews. [SCI Q1 Top, Review Article] (Submitted)
4. Lin Ruowei+, Niu Liya+, Zhao Chuang*, et al. 2025. Climate change drives escalating hot-dry wind threat to the wheat breadbasket in China. Nature Partner Journals: Sustainable Agriculture. [SCI Q1 Top] (Submitted)

CONFERENCE PARTICIPATION

- Second Youth Academic Forum of the Cropping Systems Branch, Chinese Association of Agricultural Science Societies, Lanzhou, China (2023)

SELECTED COURSEWORK

- | | | |
|------------------------------------|---------------------------------------|---|
| • Atmospheric Physics | • Cloud and Precipitation Physics | • Atmospheric Remote Sensing |
| • Atmospheric Radiation and Optics | • Modern Atmospheric Observation | • Microclimatology |
| • Air Pollution Meteorology | • Atmospheric Chemistry | • Atmospheric Turbulence |
| • Dynamic Meteorology | • Geographic Information Systems | • Statistical Analysis and Applications |
| • Agricultural Meteorology | • Agricultural and Climate Statistics | • Fundamentals of Computer Simulation |